NETWORK DESIGN & IMPLEMENTATION

# Small Office Network

VLAN Segmentation • 802.1Q Trunking • Router-on-a-Stick • Layer 2 Security

| | |
|---|---|
| **Organization** | Example Office Company (Fictional Scenario) |
| **Project Type** | Small Office LAN Design and Configuration |
| **Platform** | Cisco Packet Tracer |
| **Devices** | 1 Router • 2 Switches • 9 PCs • 1 Server |
| **Report Status** | Implementation and Verification Report |

*Prepared as a professional technical record of the proposed implementation.*

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# 1. Executive Summary

This report documents the design and configuration of a small-office local area network (LAN) for a fictional company with three departments: Human Resources (HR), Sales, and Information Technology (IT). The implementation was built and configured in Cisco Packet Tracer using one router, two Layer 2 switches, nine end-user PCs, and one server.

The network uses VLANs to separate departmental broadcast domains, IEEE 802.1Q trunk links to carry multiple VLANs over shared Ethernet links, and Router-on-a-Stick to route traffic between the three IPv4 subnets. Layer 2 hardening was also configured through sticky MAC learning, a one-MAC limit, shutdown violation mode, and administrative shutdown of unused switch ports.

The intended outcome is controlled communication between departments, predictable IP addressing, reduced unnecessary Layer 2 broadcast scope, and a configuration that can be verified through CLI commands and end-to-end tests. This report describes the intended design and configuration; final operational status should be confirmed against the actual Packet Tracer file and recorded test evidence.

# 2. Business Scenario and Requirements

The fictional organization operates three departments that share office infrastructure but require logical separation. HR handles employee records, Sales manages customer and sales operations, and IT supports internal systems and the office server. The design keeps each department in a separate VLAN while allowing required interdepartmental communication through a central router.

## 2.1 Business and technical objectives

- Create separate broadcast domains for HR, Sales, and IT.
- Provide an IPv4 address plan with a default gateway for each departmental subnet.
- Allow communication between VLANs through Router-on-a-Stick.
- Carry multiple VLANs across the inter-switch link and the router uplink using IEEE 802.1Q trunking.
- Reduce exposure from unused access ports and restrict the number of MAC addresses learned on user-facing ports.
- Provide repeatable verification steps and evidence suitable for technical handover.

## 2.2 Scope and assumptions

- The implementation is a Packet Tracer simulation, not a production deployment on physical equipment.
- The router is represented by a Cisco 2911 and the switches by Cisco 2960-24TT devices; interface availability may vary by Packet Tracer device model.
- The server is placed in VLAN 30 (IT) and uses a static IPv4 address.
- No DHCP, DNS, Internet edge, wireless access, dynamic routing, ACL policy, centralized authentication, or high availability is included in the baseline scope.
- The report does not claim that tests passed unless the result is observed and documented in the actual simulation.

# 3. Network Topology and Physical Design

R1 connects to SW1 through GigabitEthernet0/0 on the router and GigabitEthernet0/1 on SW1. SW1 connects to SW2 through GigabitEthernet0/2 on SW1 and GigabitEthernet0/1 on SW2. End devices connect to FastEthernet access ports. This layout centralizes inter-VLAN routing on R1 and extends all three departmental VLANs across both switches.

| Device | Interface | Connected Device / Port | Purpose |
|---|---|---|---|
| R1 | G0/0 | SW1 G0/1 | 802.1Q trunk uplink; Router-on-a-Stick |
| SW1 | G0/2 | SW2 G0/1 | Inter-switch trunk |
| SW1 | Fa0/1 | HR-PC1 Fa0 | VLAN 10 access |
| SW1 | Fa0/2 | HR-PC2 Fa0 | VLAN 10 access |
| SW1 | Fa0/3 | Sales-PC1 Fa0 | VLAN 20 access |
| SW1 | Fa0/4 | Sales-PC2 Fa0 | VLAN 20 access |
| SW1 | Fa0/5 | IT-PC1 Fa0 | VLAN 30 access |
| SW2 | Fa0/1 | HR-PC3 Fa0 | VLAN 10 access |
| SW2 | Fa0/2 | Sales-PC3 Fa0 | VLAN 20 access |
| SW2 | Fa0/3 | IT-PC2 Fa0 | VLAN 30 access |
| SW2 | Fa0/4 | IT-PC3 Fa0 | VLAN 30 access |
| SW2 | Fa0/5 | SRV1 Fa0 | VLAN 30 access |

Cable selection: copper straight-through Ethernet cables were used for end-device-to-switch connections and the router-to-switch link. The inter-switch link uses GigabitEthernet interfaces; Packet Tracer may also support automatic cable selection. Interface names should be checked against the actual selected device models.

# 4. Logical Segmentation and VLAN Design

VLANs segment the LAN at Layer 2. Devices in different VLANs do not communicate directly at Layer 2; traffic between them must be routed by a Layer 3 device. VLAN 99 is reserved as the native VLAN on trunk links. VLAN 999 is reserved for unused access ports and is not intended for user endpoints.

| VLAN ID | Name | Purpose | Subnet | Gateway |
|---|---|---|---|---|
| 10 | HR | Human Resources endpoints | 192.168.10.0/24 | 192.168.10.1 |
| 20 | Sales | Sales endpoints | 192.168.20.0/24 | 192.168.20.1 |
| 30 | IT | IT endpoints and server | 192.168.30.0/24 | 192.168.30.1 |
| 99 | NATIVE | Native VLAN for 802.1Q trunks; no user endpoints | No IP subnet assigned | N/A |
| 999 | UNUSED | Parking VLAN for unused switch ports | No IP subnet assigned | N/A |

The VLANs were created separately on SW1 and SW2. Creating a VLAN on one standalone switch does not automatically create it on the other unless a VLAN distribution mechanism is configured; this design uses explicit local configuration.

# 5. IPv4 Addressing Plan and Subnetting

Each departmental network uses a /24 prefix, equivalent to subnet mask 255.255.255.0. This provides 256 total addresses per subnet, including the network and broadcast addresses, leaving 254 conventional usable host addresses.

| Endpoint | VLAN | IPv4 Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| HR-PC1 | 10 | 192.168.10.11 | 255.255.255.0 | 192.168.10.1 |
| HR-PC2 | 10 | 192.168.10.12 | 255.255.255.0 | 192.168.10.1 |
| HR-PC3 | 10 | 192.168.10.13 | 255.255.255.0 | 192.168.10.1 |
| Sales-PC1 | 20 | 192.168.20.11 | 255.255.255.0 | 192.168.20.1 |

| Endpoint | VLAN | IPv4 Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| Sales-PC2 | 20 | 192.168.20.12 | 255.255.255.0 | 192.168.20.1 |
| Sales-PC3 | 20 | 192.168.20.13 | 255.255.255.0 | 192.168.20.1 |
| IT-PC1 | 30 | 192.168.30.11 | 255.255.255.0 | 192.168.30.1 |
| IT-PC2 | 30 | 192.168.30.12 | 255.255.255.0 | 192.168.30.1 |
| IT-PC3 | 30 | 192.168.30.13 | 255.255.255.0 | 192.168.30.1 |
| SRV1 | 30 | 192.168.30.100 | 255.255.255.0 | 192.168.30.1 |

All endpoint addresses are statically assigned through Desktop > IP Configuration in Packet Tracer. The gateway addresses are configured on R1 subinterfaces. The network, broadcast, and gateway addresses are not assigned to the listed PCs or server.

# 6. Switch Configuration

## 6.1 Hostnames and VLAN creation

Each switch was assigned a descriptive hostname (SW1 and SW2). VLANs 10, 20, 30, 99, and 999 were created and named HR, Sales, IT, NATIVE, and UNUSED respectively.

```
enable
configure terminal
hostname SW1

vlan 10
 name HR
exit
vlan 20
 name Sales
exit
vlan 30
 name IT
exit
vlan 99
 name NATIVE
exit
vlan 999
 name UNUSED
exit
```

The same VLAN definitions were configured on SW2 with the hostname set to SW2. Verification command: show vlan brief.

## 6.2 Access-port assignment

User-facing ports were forced into access mode and assigned to the VLAN associated with each department. This avoids relying on dynamic negotiation for ordinary endpoint ports.

```
! SW1
interface range fa0/1-2
 switchport mode access
 switchport access vlan 10
exit
interface range fa0/3-4
 switchport mode access
 switchport access vlan 20
exit
interface fa0/5
 switchport mode access
 switchport access vlan 30
exit

! SW2
interface fa0/1
 switchport mode access
 switchport access vlan 10
exit
interface fa0/2
 switchport mode access
 switchport access vlan 20
exit
interface range fa0/3-5
 switchport mode access
 switchport access vlan 30
exit
```

Verification command: show vlan brief. Expected result: the listed access ports appear under their assigned VLAN IDs.

### 6.3 802.1Q trunk configuration

Trunking is enabled on SW1 G0/1 toward R1, SW1 G0/2 toward SW2, and SW2 G0/1 toward SW1. The allowed VLAN list is explicitly limited to VLANs 10, 20, 30, and 99. VLAN 999 is not required across the trunks because it is used only to park unused local access ports.

```
! SW1 to R1
interface gigabitEthernet 0/1
 description TRUNK_TO_R1
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,99
 no shutdown
exit

! SW1 to SW2
interface gigabitEthernet 0/2
 description TRUNK_TO_SW2
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,99
 no shutdown
exit

! SW2 to SW1
interface gigabitEthernet 0/1
 description TRUNK_TO_SW1
 switchport mode trunk
 switchport trunk native vlan 99
 switchport trunk allowed vlan 10,20,30,99
 no shutdown
exit
```

The native VLAN must match on both ends of each trunk. Verification command: show interfaces trunk. On supported 2960 Packet Tracer devices, the output should show the trunking interfaces, native VLAN 99, and the permitted VLAN list.

## 7. Router-on-a-Stick Configuration

Router-on-a-Stick allows a single physical router interface to serve multiple VLANs through logical subinterfaces. Each subinterface receives an IEEE 802.1Q VLAN identifier and an IP address that serves as the default gateway for that subnet.

```
enable
configure terminal
hostname R1

interface gigabitEthernet 0/0
 no ip address
 no shutdown
exit

interface gigabitEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit

interface gigabitEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit

interface gigabitEthernet 0/0.99
 encapsulation dot1Q 99 native
exit

end
copy running-config startup-config
```

The physical G0/0 interface is enabled without a Layer 3 address; the subinterfaces carry the gateway addresses. The native VLAN subinterface has no IP address because VLAN 99 is reserved for trunk-native traffic and is not used as a client subnet in this design.

Verification commands: show ip interface brief and show ip route. The three departmental subnets should appear as directly connected routes when the interfaces and physical link are operational.

# 8. Layer 2 Security Hardening

## 8.1 Port Security with Sticky MAC

Port Security was applied to access ports Fa0/1–Fa0/5 on both switches. Each protected port allows one secure MAC address, learns that address through sticky learning, and uses shutdown as the violation action.

```
interface range fa0/1-5
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
exit
```

Sticky learning records dynamically learned secure MAC addresses in the running configuration. To retain these learned entries after a reload, the configuration must be saved after the addresses have been learned. A different MAC address appearing on a protected port can trigger a violation and place the port into a secure-shutdown/error-disabled state, depending on platform behavior.

Verification commands: show port-security, show port-security interface fa0/1, and show port-security address. Generate traffic first if no sticky address has yet been learned.

## 8.2 Unused-port shutdown

Unused FastEthernet ports Fa0/6–Fa0/24 on both switches were assigned to VLAN 999 and administratively disabled. SW2 G0/2 was also disabled because it is not used in the documented topology.

```
! On both SW1 and SW2
interface range fa0/6-24
 switchport mode access
 switchport access vlan 999
 shutdown
exit

! On SW2 only
interface gigabitEthernet 0/2
 switchport mode access
 switchport access vlan 999
 shutdown
exit
```

This practice reduces the risk of an unauthorized device being connected to an unused port. Verification command: show interfaces status. Confirm that active uplinks and user ports have not been included in the shutdown range.

# 9. MAC Address Learning and Switching Behavior

The switches dynamically learn source MAC addresses from frames received on their interfaces. The learned entries are associated with a VLAN and an interface. Within a VLAN, known unicast traffic is forwarded based on the MAC address table; unknown unicast and broadcast behavior is scoped to the VLAN, subject to normal switching rules.

```
show mac address-table dynamic
show mac address-table dynamic vlan 10
```

To populate the table, a test ping can be initiated between two hosts in the same VLAN. A switch learns the source MAC from the traffic it receives; the initial ARP exchange may also generate entries and additional frames.

# 10. Verification and Acceptance Test Plan

The following test plan defines the checks required to accept the implementation. The actual result for each test should be recorded after running it in the Packet Tracer file; the table intentionally does not assume success without captured evidence.

| Test ID | Test / Command | Expected Result | Observed Result |
|---|---|---|---|
| T01 | SW1/SW2: show vlan brief | VLANs 10, 20, 30, 99, 999 exist; access ports assigned correctly. | To be recorded |
| T02 | SW1/SW2: show interfaces trunk | Expected uplinks are trunking; native VLAN 99; allowed VLANs correct. | To be recorded |
| T03 | R1: show ip interface brief | G0/0 and configured subinterfaces operational (up/up when links are active). | To be recorded |
| T04 | R1: show ip route | Connected routes for 192.168.10.0/24, .20.0/24, and .30.0/24. | To be recorded |
| T05 | HR-PC1 ping 192.168.10.12 | Successful intra-VLAN connectivity after ARP resolution. | To be recorded |
| T06 | HR-PC1 ping 192.168.10.1 | HR host can reach its default gateway. | To be recorded |
| T07 | HR-PC1 ping 192.168.20.11 | Successful inter-VLAN connectivity via R1. | To be recorded |
| T08 | HR-PC1 ping 192.168.30.100 | HR host can reach the server through inter-VLAN routing. | To be recorded |
| T09 | SW1: show mac address-table dynamic | Learned MAC entries appear against the correct VLAN and interfaces after traffic. | To be recorded |
| T10 | SW1/SW2: show port-security interface fa0/1 | Port Security enabled; maximum one secure MAC; shutdown violation mode. | To be recorded |
| T11 | SW1/SW2: show interfaces status | Unused ports are disabled; production links remain active. | To be recorded |
| T12 | Packet Tracer Simulation Mode: HR-PC1 to Sales-PC1 | Event path shows the packet reaching R1 for inter-VLAN routing. | To be recorded |

# 11. Demonstrating Inter-VLAN Routing

A ping from HR-PC1 (192.168.10.11) to Sales-PC1 (192.168.20.11) is an inter-subnet communication test. HR-PC1 identifies that the destination is outside its local /24 subnet and sends the Ethernet frame to its default gateway, 192.168.10.1. SW1 forwards the VLAN 10 frame across the trunk to R1. R1 routes the IP packet using its connected route for 192.168.20.0/24 and sends a new Ethernet frame tagged for VLAN 20. The switching infrastructure then delivers it to Sales-PC1.

The return packet follows the reverse routed path through R1. The IP source and destination addresses remain the end hosts' addresses, while the Ethernet source and destination MAC addresses are rewritten at the routed hop. Use Simulation Mode and Capture/Forward to observe the path and distinguish ARP events from ICMP events.

A successful ping alone confirms reachability, not the complete path. The simulation event sequence and the router's connected routes provide stronger evidence that the communication uses the router.

# 12. Configuration Management and Handover

- Save the running configuration on R1, SW1, and SW2 using copy running-config startup-config after validating the configuration.
- Save the Packet Tracer project as Small_Office_Network.pkt.
- Capture evidence of VLAN membership, trunk status, router interfaces and routes, MAC learning, Port Security, unused-port status, and successful connectivity tests.
- Record the actual test outcomes and any exceptions before signing off the implementation.
- Keep a copy of the addressing table and port map with the project file for future troubleshooting.

# 13. Limitations and Recommendations

This is a learning-scale, single-router design. It demonstrates core Layer 2 segmentation and inter-VLAN routing but does not implement a full enterprise security policy.

- Add Access Control Lists (ACLs) on the router if HR, Sales, and IT require different access permissions; VLAN separation alone does not prohibit routing between departments.
- Consider DHCP scopes per VLAN if endpoint counts grow, while reserving static addresses for network infrastructure and servers.
- Use a dedicated management VLAN and secure management protocols such as SSH for switch/router administration in a production design.
- Consider DHCP snooping, Dynamic ARP Inspection, BPDU Guard, and PortFast on appropriate edge ports where supported and designed safely.
- Evaluate whether a Layer 3 switch, redundant uplinks, and first-hop redundancy are required as the organization grows.
- Use documented change control and test security behavior before deployment to physical production equipment.

# 14. Conclusion

The proposed Small Office Network establishes a structured baseline for a three-department office. VLANs 10, 20, and 30 separate HR, Sales, and IT; 802.1Q trunks extend those VLANs across the switching infrastructure; and Router-on-a-Stick provides gateways and routing between the departmental IPv4 networks. Sticky MAC Port Security, a one-address limit, shutdown violation mode, and shutdown of unused ports add basic Layer 2 hardening.

The implementation should be considered complete only after the verification plan is executed, observed results are recorded, and the Packet Tracer project and supporting screenshots are saved. This distinction between configured intent and verified outcome is essential in a professional network handover.

# Appendix A — Quick Verification Commands

```
! Switches
show vlan brief
show interfaces trunk
show mac address-table dynamic
show port-security
show port-security interface fa0/1
show port-security address
show interfaces status
show running-config

! Router
show ip interface brief
show ip route
show ip arp
show running-config

! PC command prompt
ipconfig
ping 192.168.10.1
ping 192.168.10.12
ping 192.168.20.11
ping 192.168.30.100
```

# Appendix B — Implementation Notes

The report is based on the stated Packet Tracer design and configuration plan. Device model capabilities, interface names, and exact CLI output can vary slightly by Packet Tracer version. Where a verification outcome is not yet captured, it is explicitly marked “To be recorded” rather than presented as a confirmed result.

Prepared for educational and technical documentation purposes. The company scenario is fictional and does not represent a real organization's confidential network.